Example 4.17 Simplify $\frac{7!}{2!}$

Solution:

$$\frac{7!}{2!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2!}{2!} = 7 \times 6 \times 5 \times 4 \times 3 = 2520.$$

Example 4.18 Evaluate $\frac{n!}{r!(n-r)!}$ when (i) $n = 7, r = 5$ (ii) $n = 50, r = 47$ (iii) For any n with $r = 3$.

Solution:

(i) When $n = 7, r = 5$

$$\frac{n!}{r!(n-r)!} = \frac{7!}{5!(7-5)!} = \frac{7 \times 6 \times 5!}{5! \times 2!} = \frac{7 \times 6}{1 \times 2} = 21.$$

(ii) When $n = 50, r = 47$

$$\frac{n!}{r!(n-r)!} = \frac{50!}{47!(50-47)!} = \frac{50 \times 49 \times 48 \times 47!}{47! \times 3!} = \frac{50 \times 49 \times 48}{1 \times 2 \times 3} = 19600.$$

(iii) For any n and $r = 3$

$$\frac{n!}{r!(n-r)!} = \frac{n!}{3!(n-3)!} = \frac{n \times (n-1) \times (n-2)(n-3)!}{1 \times 2 \times 3 \times (n-3)!} = \frac{n(n-1)(n-2)}{6}.$$

Example 4.19 Let N denote the number of days. If the value of $N!$ is equal to the total number of hours in N days then find the value of N?

Solution:
We need to solve the equation $N! = 24 \times N$.
For $N = 1, 2, 3, 4, \quad N! < 24 \times N$.
For $N = 5$, we have $N! = 5! = 4! \times 5 = 24N$.
For $N > 5$, we have $N! \geq 5!N > 24 \times N$. Hence $N = 5$.

Example 4.20 If $\frac{6!}{n!} = 6$, then find the value of n.

Solution:

$\frac{6!}{n!} = \frac{1.2.3.4.5.6.}{1.2.3...n} = 6$. As $n < 6$ we get, $n = 5$.

Example 4.21 If $n! + (n-1)! = 30$, then find the value of n.

Solution:
Now, $30 = 6 \times 5$. As $n! + (n-1)! = (n+1)(n-1)!$, equating $(n-1)! = 6 = 3!$, we get $n = 4$.